

Ben Bubnick

DATA ENGINEER · DATA SCIENTIST · ANALYTICS SPECIALIST

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Summary

Hands-on engineer specializing in Python on UNIX-like systems (Linux/macOS), production ML/MLOps, and large-scale data pipelines. Heavy daily use of Terminal/Bash/Zsh, SSH, and POSIX tooling to build and operate services. Built forecasting and optimization solutions (e.g., surgery duration prediction, hospitalization risk) and led end-to-end data platforms in AWS and Azure. Strong at turning ambiguous operational problems into real-time services with CI/CD, observability, and documentation. Comfortable collaborating across engineering, data science, and ops—ready to lead AI application development and model integration in high-performance, resource- and cost-sensitive environments.

Experience

GrowData Analytics

AWS: S3, Glue, Athena, Lambda, CloudWatch, Ruby, RubyLLM, Tesseract OCR, Rubocop, RSpec

PRINCIPAL ML ENGINEER

Sep. 2024 - Present

Built and shipped an OCR-to-analytics product for small farms; owned model integration, CI/CD, testing, and auditability on UNIX-like systems.

- Productized OCR → structured data pipeline for farm invoices (handwritten/printed) using Tesseract + rule-based post-processing; shipped as a small Python/Ruby service with containerized deployments and versioned configs.
- Built validation & reconciliation layers (schema checks, cross-field rules, idempotent re-runs) and produced audit logs to support bookkeeping/compliance workflows.
- Implemented CI/CD (unit tests, linting, fixtures, golden-file tests) and release automation; documented runbooks and failure modes (timeouts, low-confidence OCR, retries).
- Delivered a lightweight analytics layer (usage metrics, error taxonomy, latency/throughput dashboards) to guide feature tuning and cost control.
- Owned infra as code for a minimal AWS footprint (S3 for artifacts, Lambda/Cron for jobs, CloudWatch alerts); operated via UNIX-like CLI workflows (Bash/Zsh, Makefile targets).

Health Data Movers

AWS: S3, Snowflake, SQLMesh, Python, SQL, Docker

DATA CONSULTANT

Mar. 2025 - Present

Developed a reusable ETL pipeline to streamline data onboarding for a healthcare revenue cycle platform.

- Built a reusable ingestion & transformation framework in SQLMesh/Snowflake (templated models, data contracts, CI checks, promotion gates), enabling consistent, repeatable onboardings.
- Added observability & auditability (row-count/parity checks, schema drift alerts, reconciliation logs) and wrote runbooks for deterministic re-runs and incident handling on UNIX-like systems.
- Shipped a lightweight forecasting/alerting pattern (Python service + Docker + Actions) for operations signals; packaged with tests and deployment automation.

Arkos Health

AWS: S3, Athena, Redshift, Glue, dbt, PySpark, Docker + AWS CLI, Tableau, Jupyter

SENIOR DATA ANALYST

Jun. 2023 - Jan. 2025

Built a scalable AWS data lake with automated CI/CD, metadata reporting, and integrated analytics tools.

- Stood up an AWS data lake with CI/CD (dbt + Git flows), cost-aware partitioning, and job scheduling; reduced deployment time 50% and improved query performance with targeted model refactors.
- Implemented monitoring/alerting & on-call runbooks (metrics, logs, failure taxonomies) to lower MTTR and improve data reliability.
- Mentored engineers on code reviews, testing, and pipeline design; authored internal standards that lifted team delivery quality.

Merative

Azure: Data Factory, Blob Storage, Functions, SQL DB, Synapse Analytics, Python, DevOps

SENIOR DATA SCIENTIST

Jul. 2022 - Mar. 2023

Launched a new Azure platform to replace HDFS systems to modernize data infrastructure.

- Led HIPAA-compliant deidentification initiative across 200+ million patient records.
- Designed and implemented an automated testing framework for ETL pipelines for validation coverage.
- Validated and optimized Azure-based data workflows, ensuring continuity and fidelity during the transition from legacy HDFS systems to cloud architecture.
- Conducted end-to-end system testing for Azure Data Factory pipelines, Blob Storage configurations, and downstream Synapse integrations.
- Collaborated with engineering teams across disciplines to refine architectural designs.
- Documented test plans and platform migration procedures for improving onboarding for future QA engineers.
- Built regression test suites and data comparison tools, detecting schema drift and transformation errors early in the development lifecycle.
- Ensured compliance with data governance and security protocols to reduce audit risks and support HITRUST readiness.

IBM

Hadoop/HDFS/CDH: Spark, Hive & Impala, Kafka, Python, JRuby, SQL, Pig, Docker + Gradle

SENIOR DATA SCIENTIST

Nov. 2021 - Jul. 2022

Built AI tools while leading analytics delivery and large-scale data validation efforts.

- Created AI Surgical Scheduling Optimization tool currently evolving into new offering.
- Led delivery on tailored analytics and reporting project for client administrative data.
- Developed data migration validation analysis on 150+ billion patient records.
- Developed surgical smart case card using combined CNN/RNN model.

SENIOR DATA CURATION ARCHITECT

Dec. 2020 - Nov. 2021

Led large-scale clinical data and NLP initiatives and delivering ML solutions.

- Developed clinical data integration of 100+ billion structured & semi-structured records.
- Developed topic modelling project on nearly 6 million semi-structured provider notes data.
- Curated 25 pages of workflow documentation and mentored new hires in AI tuning process.
- Created COVID-19 hospitalization prediction tool with Machine Learning algorithms.

DATA CURATION ARCHITECT

Nov. 2017 - Dec. 2020

Led enterprise data integration initiatives and mentoring programs.

- Managed multi-org enterprise integration with over 70 interconnected data sources.
- Developed new data validation and review process that reduced total rework by 25%.
- Reduced mapping rework by 45% with through new knowledge-driven initiatives.
- Led mentoring program to increase 4,500 billable hours over two year period.

SOFTWARE DEVELOPER

Sep. 2016 - Nov. 2017

Developed internal resources that reduced costs and improved onboarding efficiency.

- Reduced time spent on knowledge transfer by 80% by rewriting project documentation.
- Eliminated need for a \$30,000 annual contract by developing data curation tool.
- Reduced cost-per-hire by \$5,210 by re-developing education program.

DATA SCIENTIST

Nov. 2015 - Sep. 2016

Led integration efforts and process automation initiatives.

- Managed team of 12 remote data science contractors for 36 integration projects.
- Created over 125 new knowledge base articles, making up 30% of the team's online content.
- Developed automated data integration tool to cut out 28% unnecessary development time.

Explorys

Hadoop/HDFS: Ruby, Pig, React, NodeJS, JavaScript, SQL, Oracle, Jenkins

DATA SCIENTIST

Jun. 2015 - Nov. 2015

Built ETL pipelines on distributed data systems to support clinical data integration and population analytics.

- Developed cron-based widget to define blackout periods for client data pulls.
- Performed troubleshooting on production issues across multiple environments.
- Principal integration specialist for 10+ client database warehouses.

Other Experience

ADJUNCT PROFESSOR, **Multiple Institutions**, *Aug. 2010 - May. 2019.*
SOFTWARE ENGINEER, **AmTrust Financial Services**, *Mar. 2014 - Jun. 2015.*
COLOR SCIENTIST, **PPG Industries**, *Apr. 2012 - Jun. 2013.*

Education

2022	PGCert. , Data Science Professional	IBM
2015	PGCert. , Data Science	Johns Hopkins University
2010	Master of Science , Astrophysics	University of Cincinnati
2004	Bachelor of Fine Arts , Printmaking	Ohio University